Math Warning! △

Only necessary formulae will be introduced!

- no multi-view — ELBO, Jensen
- no association — VAE → EM

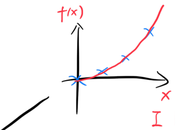
Text to Image

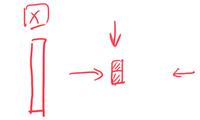
Basic of Generation

Understanding: $P(y|x)$ $\tilde{x} \to y$

- Task: Estimate $P(x)$ and Sample $\tilde{x}$ from learned $F(\cdot)$
- Difficulty: Limited data points. – Predict unseen density
- Solution: Add Prior – constraints on search space.

curse of dimensionality 10000^{10}

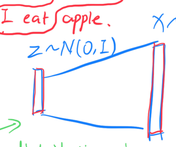




GPT ?

I like apple.
I like orange.
I eat apple.

GAN



$D = \{x_1, x_2, \cdots, x_n\}$

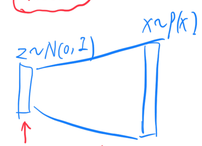
$\to \{z_1, z_2, z_3, \cdots, z_n\}$ KL

$\to \{\tilde{x}_1, \tilde{x}_2, \tilde{x}_3, \cdots, \tilde{x}_n\}$

whole distribution view

Basic of VAE

VAE



$D = \{x_1, x_2, \cdots, x_n\}$

i.i.d

single point view

$P(D) = \prod p(x_i)$

$\sum_{i=1}^{n} \log p(x_i)$

$p(x|z) \sim N(F(z), I)$

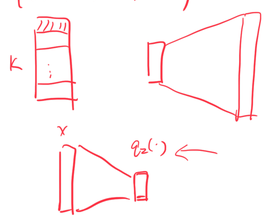
VQ-VAE

CogView

$p(z_{i+1} | z_0 z_1 z_2 \cdots z_i)$

Z K $\{0, 1, 2, \cdots, k-1\}$

$P_z(\cdot)$



ELBO $p(x) \geq -KL(q_z(\cdot) \| P_{x,z}(x,\cdot))$

$\log p(x) = \log \int_z p(x,z)\, dz$

$= \log \int_z \frac{p(x,z)}{q(z)} q(z)\, dz$

$= \log E_{z\sim q(z)}\left[\frac{p(x,z)}{q(z)}\right]$

$\geq E_{z\sim q(z)}\left[\log \frac{p(x,z)}{q(z)}\right]$ △

$= E_{z\sim q(z)}\left[\log \frac{p(x|z)\,p(z)}{q(z)}\right]$

$\int f(x)\, p(x)\, dx = E_{x\sim p(x)}[f(x)]$ △

$F(E[X]) \geq E[F(x)]$

$E_{z\sim q(z)}\left[\log \frac{q(z)}{p(z)}\right] = \int q(z) \cdot \log \frac{q(z)}{p(z)}\, dz$

$= KL(q_z(\cdot) \| p_z(\cdot))$

$KL(N(u,\sigma) \| N(0,1)) = -\frac{1}{2}(1 + \log\sigma^2 - u^2 - \sigma^2)$

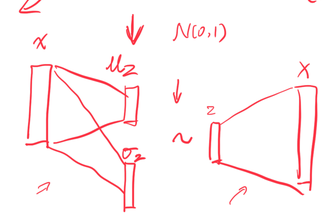
$= E_{z\sim q(z)}[\log p(x|z)] - E_{z\sim q(z)}\left[\log \frac{q(z)}{p(z)}\right]$

$= E_{z\sim q(z)}[\log p(x|z)] - KL(q_z(\cdot) \| p_z(\cdot))$

$p(x|z) = \frac{1}{(\sqrt{2\pi})^d} e^{-\frac{\|x-F(z)\|^2}{2}}$

$\log p(x|z) = \log \frac{1}{(\sqrt{2\pi})^d} + \log e^{-\frac{\|x-F(z)\|^2}{2}}$

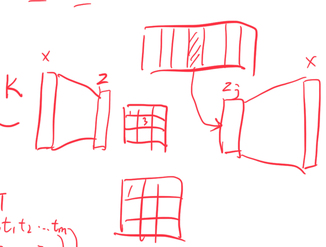
$\Rightarrow -\frac{\|x-F(z)\|^2}{2}$



$ELBO_{VAE}(x) = E_{z\sim q(z)}\left[-\frac{\|x-F(z)\|^2}{2}\right] + \frac{1}{2}\sum_{j=1}^{d}(1 + \log\sigma_j^2 - u_j^2 - \sigma_j^2)$

$ELBO_{VQ\text{-}VAE}(x) = E_{z\sim q(z)}\left[-\frac{\|x-F(z)\|^2}{2}\right] - \log K$

$\Rightarrow -\frac{\|x-F(z_j)\|^2}{2}$



$KL(q_z(\cdot) \| p(\cdot | z_0 z_1 \cdots z_i))$ GPT $t_0 t_1 t_2 \cdots t_m$

$= -\log p(z_j | z_0 z_1 \cdots z_i)$

标题	年
Conditional Image Generation with PixelCNN Decoders	2016
Zero-Shot Text-to-Image Generation	2021
An Introduction to Variational Autoencoders	2019
Neural Discrete Representation Learning	2018
The Road from MLE to EM to VAE: A Brief Tutorial	2020
CogView: Mastering Text-to-Image Generation via Tran…	2021